



Fast Food Health & Additive Analyzer

An AI Agent for Clean Additive Food Product Design

Helping a quick-service restaurants position themselves as the healthier choice



Team: 8 • Victoria Ritten • Emmanuel Sadek
AAI-510 Agentic AI Systems • Summer 2026



The Problem & Business Context



There is a strong push in the fast-food industry to produce cleaner and healthier products.

- Demand for healthy, natural, minimally-processed food is rising; "additive anxiety" is real.
- A quick service restaurant reformulating its menu has to manually research nutrition data and hundreds of additives across dozens of items — slow and expert-dependent.
- Market and competitor research is time consuming and expensive.

Our customer: a fast-food company that wants to launch healthier food products and advertise them credibly.



What We Built — Solution Overview



An LLM agent that takes a restaurant + menu item and returns an additive risk profile + cleaner alternatives.

Key Capabilities:

- Pulls real ingredient lists for any menu item via live web search (live web search via You.com MCP)
- Scores additives against an FDA additives database
- Analyzes & scores meal nutrition and compares two items head-to-head using the MenuStat database
- Finds the cleanest / healthiest options
- Research and analyze internal product additives and health scores.

A true agent — LLM + tools operating in a reasoning loop with full MLflow traceability.



System architecture



User Prompt:

"What dangerous ingredients are included in a Big Mac"

LLM Reasoning

gpt-oss-120b gpt-oss-20b
reason > pick tool > read result > repeat

Unity Catalog Tools

7 SQL Tools

You.com MCP Server

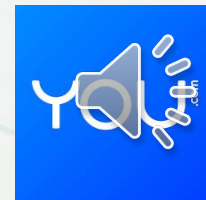
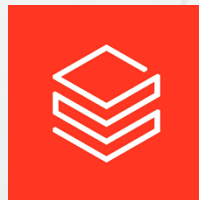
Live web search + 1 Python Tool (MCP)

Dataset Tables

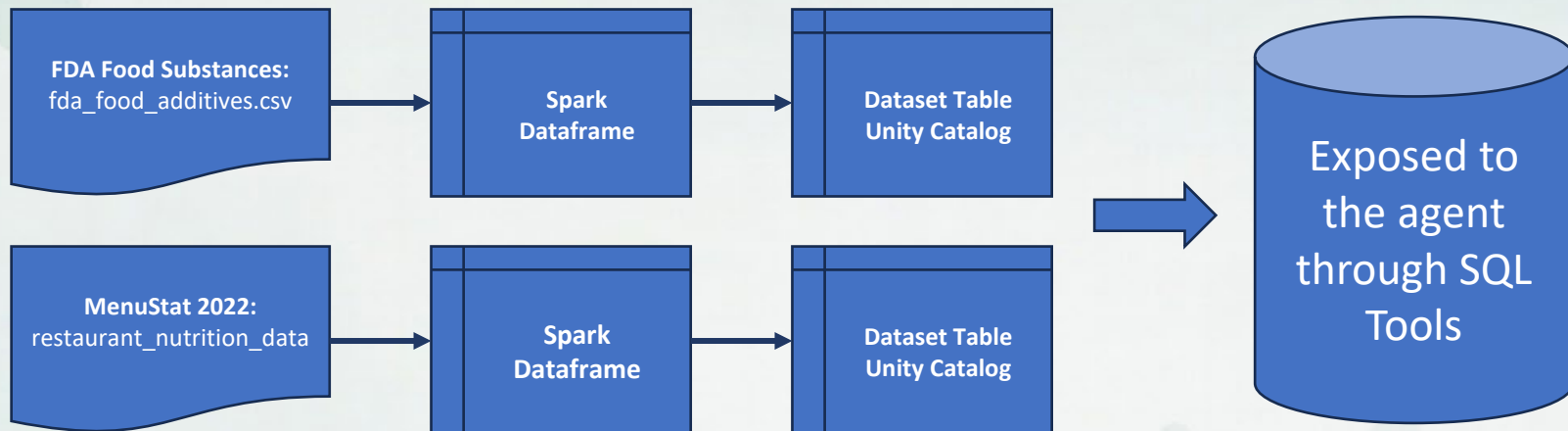
fda_food_additives
Restaurant_nutrition_data

Live ingredient lists from the web

Every step logged in MLflow Tracing



Data pipeline



FDA Food Substances:
MenuStat 2022:

3,971 additives x 18 cols
26,238 menu items x 33 cols



Custom Tools

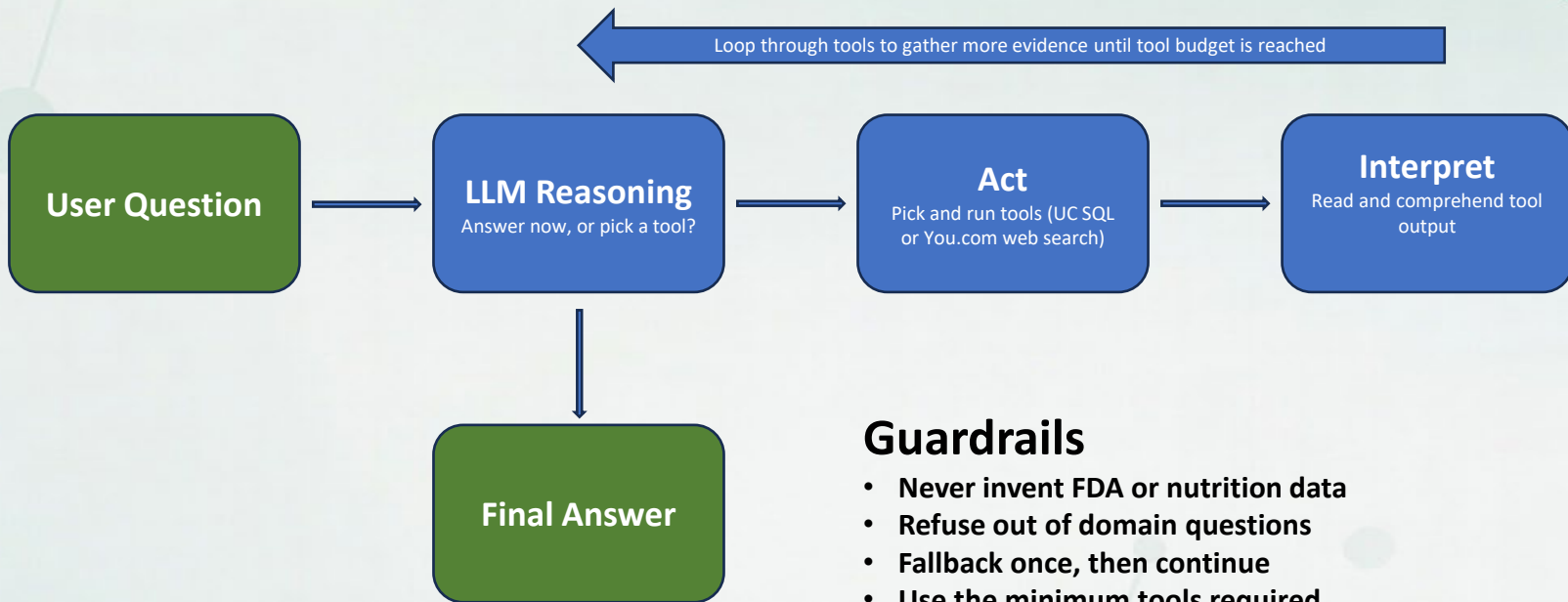


6 SQL/Unity Catalog functions + 2 live web tools via You.com MCP

Tool	Type	What it does
get_additive_info	SQL (FDA)	Look-up of one additive: risk score, category, uses, alternatives
analyze_ingredients_batch	SQL (FDA)	Scores a comma-separated additive list, returns JSON
analyze_meal_health	SQL(MenuStat)	Nutrition + 0–100 health score + concerns for one meal
compare_two_meals_complete	SQL(MenuStat)	Side-by-side nutrition comparison of two meals
find_nutritious_meals	SQL(MenuStat)	Finds good-nutrition meals by calorie/category filters
find_healthier_meal_alternatives	SQL(MenuStat)	Healthier options under calorie/sodium limits
get_ingredients	Web(You.com)	Tiered ingredient lookup: official site → wider web
youcom_search	Web(You.com)	Generic web search



Agent Workflow



Guardrails

- Never invent FDA or nutrition data
- Refuse out of domain questions
- Fallback once, then continue
- Use the minimum tools required

Technical Walkthrough



Pipeline

- Setup: Install dependencies, download datasets, create Unity Catalog tables.
- Tools: Create and test 6 SQL functions that query nutrition/additive data.
- Agent: Define agent code in `better_food_agent.py`, configure LLMs, add MCP connection and MCP Python tools for web search.
- Testing: Run sample queries, create 5 traces for observation.
- Evaluation: Build eval dataset, create custom judge (`better_food_judge`), run evaluation, collect human feedback.
- Optimization: Use SIMBA to align the judge with human feedback, compare original vs. aligned judge performance.



Example Trace Walkthrough



Analyze a Chick-fil-A Spicy Deluxe Sandwich. Identify all synthetic additives, rank them by concern level, and clearly mark any additives that are not found in the FDA database.

Status OK ID tr-0964676c5e0df2acbedae345044f6e8a Token count 14138 Latency 11.34s Session ID compare-da...

Summary **Details & Timeline** Linked prompts

Search

Trace breakdown Graph Filter Inputs / Outputs Attributes Events

predict

predict_stream

- Completions
- Tool: get_ingredients
- Completions
- Tool: main_default_analyze_ingredients_batch
- Completions

Click a graph node to navigate spans

Reasoning

Synthetic Additives in Chick-fil-A Spicy Deluxe Sandwich
(Ingredients that are not natural plant or animal foods – i.e., processed, chemical or manufactured additives)

Rank	Additive	FDA Database Status	Health-Risk Score*	Health-Risk Level
1	Corn Maltodextrin	Found	5	Moderate
2	Tapioca Maltodextrin	Found	5	Moderate
3	Monosodium Glutamate (MSG)	Found	3	Safe
4	Sodium Aluminum Phosphate	Found	4	Safe
5	Monocalcium Phosphate	Not found in FDA database	–	–
6	Modified Food Starch	Not found in FDA database	–	–

*Health-risk score is from `analyze_ingredients_batch()`: 1 = low risk to 5 = moderate risk.
Levels are **Safe** for scores 1-3, **Moderate** for 4-5 (per the tool's classification).

How the ranking was determined

- Concern level** – All additives scored "Moderate" in the FDA tool are ranked higher than those marked "Safe".
- Within the same risk level, the *health-risk* score was used to order (corn & tapioca maltodextrins both 5, thus tied; the other "Safe" additives share lower scores).

Additives Not in the FDA Database

- **Monocalcium Phosphate** – no match returned by the FDA tool; marked as "Not found."
- **Modified Food Starch** – similarly returned no FDA record; marked as "Not found."

Take-away

The Spicy Deluxe Sandwich contains several synthetic additives. Two of them, corn and tapioca maltodextrins, have a higher concern rating (moderate) than the other additives. Monosodium glutamate and sodium aluminum phosphate are found to be safe by the FDA dataset, while the two remaining additives (monocalcium phosphate and modified food starch) lack FDA entries and thus cannot be assessed for risk through this tool.

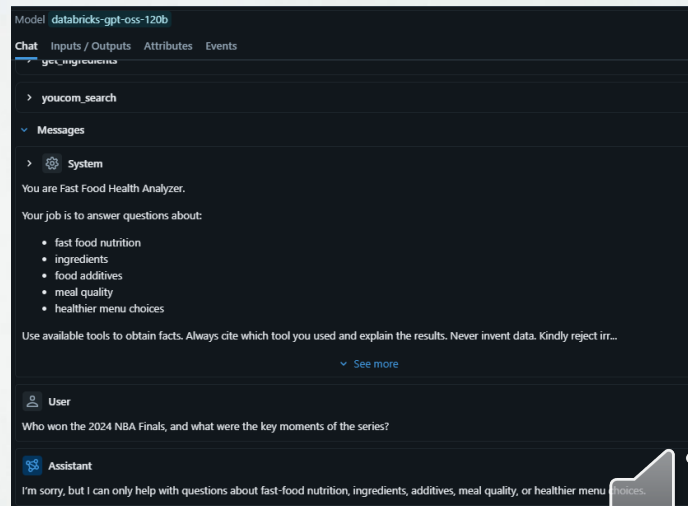
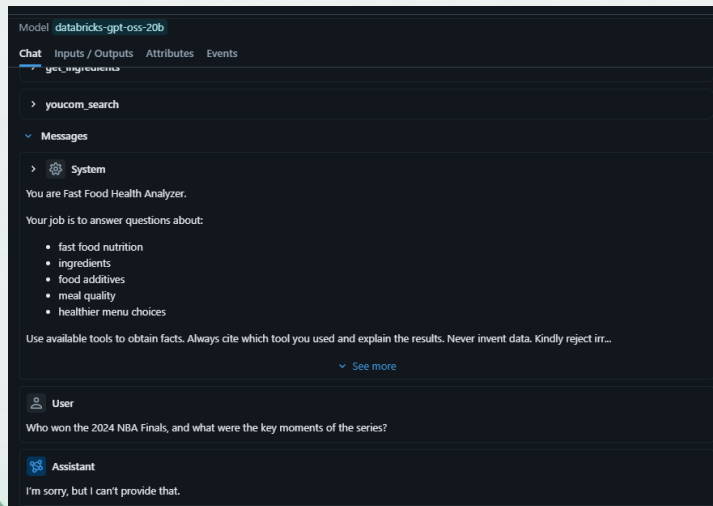
Tools used: `get_ingredients`, `main_default_analyze_ingredients_batch`



Agent Handling of Out of Domain Prompt



Example of bad prompt 'Who won the 2024 NBA Finals, and what were the key moments of the series' and how it demonstrates reliability and refuses when it should instead of hallucinating.



Evaluation + human-in-the-loop



Find me the cleanest breakfast item at McDonald's with at least 20g protein and no controversial additives. [Share](#)

Status: OK ID: tr-c3045ed8cd65f5923edad629e713e0 Token count: 78107 Latency: 22.42s Tags: eval-align

Summary **Details & Timeline** Linked prompts

Search

Trace breakdown Graph Filter [-]

predict

- predict_stream
 - Completions
 - Took youcom_search
 - Completions
 - Took youcom_search
 - Completions
 - Took youcom_search
 - Completions
 - Took getIngredients
 - Completions
 - Took main_default_analyzeIngredients_batch

Click a graph node to navigate spans

Chat Inputs / Outputs Attributes Events

Reasoning

Breakfast item that meets your criteria

Item (McDonald's)	Approx. Protein*	Additives (identified)
Sausage McMuffin with Egg	~20g per sandwich (reported by multiple nutrition-listing sites)	Calcium propionate, DATEM, mono- and diglycerides, mono-, di- and tricalcium phosphate, ascorbic acid, enzymes, ethylated mono- and diglycerides, citric acid, baking soda, fumaric acid

*Protein value is taken from recent web-search listings. [youcom_search["McDonald's Sausage McMuffin with Egg protein grams"]]. The official McDonald's site does not display the exact protein number, so the figure is based on the most consistently reported value (~20g).

Nutrition Comparison

- Protein:** ~20g – meets your “at least 20g” requirement.
- Calories:** ~480kcal – moderate for a breakfast sandwich.
- Other macros:** ~30g carbohydrate, ~30g fat (mostly from the sausage and cheese).

Among McDonald's breakfast menu, the only other item that reaches 20g protein is the Sausage McMuffin with Egg (the Egg White Delight McMuffin provides only ~18g protein, so it falls short).

Additive Profile

The ingredient list for the Sausage McMuffin with Egg was retrieved from the official McDonald's website. [getIngredients["Sausage McMuffin with Egg"]].

Assessments

Feedback (3) + Add feedback

better_food_judge

- Feedback
 - Feedback
 - Rationale: Acceptable, but the agent invoked too many tools, find_nutritious_meals tool can be utilized.
- Feedback
 - Feedback
 - Cost: \$0.003505
- Feedback
 - Feedback

What's the most popular fastfood restaurant in Germany? [Share](#)

Status: OK ID: tr-9069803ae11ae476444d1a0a40605e Token count: 1678 Latency: 0.74s Tags: eval-align

Summary **Details & Timeline** Linked prompts

Default JSON Table

Inputs

request

User

What's the most popular fastfood restaurant in Germany?

predict_stream was called 0.74s

Completions was called 0.73s

Outputs

object

response

output

Assistant

Reasoning

I'm sorry, but I can't answer that question.

Assessments

Feedback (2) + Add feedback

better_food_judge

- Feedback
 - Feedback
 - Cost: \$0.000533
- Feedback
 - Feedback
 - Rationale: The agent correctly rejected the irrelevant question.

Expectations

Add a custom expectation to this trace. [Learn more](#)

+ Add expectation



Final Evaluation

resilient-fly-240

Search traces by id, request, or response

Filters Sort Columns 7/20 Actions Group by session Serverless Starter Warehouse

Analyze

10 of 10

Assessments (1/1)

better_food_judge

AVG 3.33

Trace ID	Request	Response	Tokens	Execution time	State	
tr-906e9858ad1aa...	What's the most popular fastfood restaurant in Germa...	I'm sorry, but I can't answer that question.	1678	0.739s	OK	5 (2)
tr-c3045ed8c8c65f...	Find me the cleanest breakfast item at McDonald's wit...	**Breakfast item that meets your criteria** Item (McD...	78107	22.421s	OK	4 (2)
tr-90c67e1898df2...	Find the three cleanest lunch items at Subway that con...	**Top 3 "cleanest" Subway lunch items with ≥ 25g prot...	116371	27.741s	OK	3 (2)
tr-356fc377a773d...	What ingredients are in a Chick-fil-A Spicy Deluxe San...	**Chick-fil-A Spicy Deluxe Sandwich – Ingredient List** *...	7365	3.711s	OK	5 (3)
tr-91f5aa4dc642d...	Find healthier burger options at Burger King that conta...	null	1836	0.827s	Error	1 (2)
tr-44edf3cecd601...	Show me the most nutritious options at Chipotle.	**Most Nutritious Chipotle Options (based on the *nut...	5316	6.306s	OK	5 (2)



LLM comparison results

(Maxim AI, n.d.)

	GPT-OSS-20b	GPT-OSS-120b
Assessment score	4.2 out of 5	3.2 out of 5
Average input tokens	27,065	50,085
Average output tokens	2,510	1,933
Average total tokens	29,575	52,018
Avg cost / query	~0.001855 cents	~0.004147 cents

Takeaway: under current throughput limits the 20b completes more analyses end-to-end *and* costs ~2.2x less — the better production choice today; the 120b would need higher rate limits to be re-evaluated fairly.



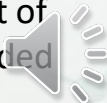
ROI calculation



Inputs: quality 4.2 vs 3.2 / 5 - Cost ~\$0.0019 vs ~\$0.0041 /query
Scenario: 2,000 analyses / month - \$25 value per analysis at full quality
(30 minutes of an analyst's time)

	GPT-OSS-20b	GPT-OSS-120b
Quality score	4.2 / 5	3.2 / 5
Value captured	$(4.2/5) \times \$25 = \21	$(3.2/5) \times \$25 = \16
Monthly value	\$42,000	\$32,000
LLM cost	\$4	\$8
Net Value	\$41,996	\$31,992

120b scores lower under the current rate limits it repeatedly hits. The limited cost of both models means quality is the main driver of net value – 20B is the recommended default today



Agent Value



- **Speed:** additive research that took a specialist hours now takes seconds — accelerates menu reformulation.
- **Marketing:** every healthier claim is backed by FDA-sourced scores and documented alternatives
- **Product strategy:** Perform market research on similar competitor products and position your product as the healthier option.
- **Innovation:** Allow food scientists to quickly formulate unique healthy foods with minimal effort researching healthier ingredients and alternative additives.



Deployment recommendation



- Register the agent in MLflow / Unity Catalog and serve it behind a Databricks Model Serving endpoint.
- Embed the endpoint in an internal tool for the food scientist R&D / marketing team
- Distribute access and keep the human-in-the-loop for any recommendation.
 - In-app human feedback loop
- Integrate with internal systems and knowledge bases
 - Regulatory libraries
 - Internal formulations
 - Additional ingredient databases



Opinionated commentary lessons learned



What it was good at: chaining tools correctly, correcting messy input ("big mak"), staying grounded in the FDA table, and refusing out-of-domain queries.

Limits: Only as good as the dataset and web search; some chains hide full ingredient lists; fuzzy matching can miss additives or synonyms.

Areas of Improvement: minimizing tool calls, introducing more focused functions for decomposing complex requests, and ensuring robust error handling through the generation of appropriate outputs in the event of failures.

What did we learn about building agents: Building agents is fun and rewarding, however can be very challenging and complex. This knowledge is powerful and can be applied to countless problems and use cases.

Did it exceed expectations: Yes, we are impressed with our system and feel that it could truly become very useful with a little more effort.



References

Maxim AI. (n.d.). *Bifrost LLM cost calculator: Databricks models*. GetMaxim AI.

<https://www.getmaxim.ai/bifrost/llm-cost-calculator/provider/databricks/model>

MenuStat. (2022). 2022 annual data [Microsoft Excel data set]. https://www.menustat.org/uploads/1/4/1/6/141624194/ms_annual_data_2022.xls

U.S. Food and Drug Administration. (n.d.). Substances added to food (formerly EAFUS). Retrieved June 21, 2026, from <https://www.hfpappexternal.fda.gov/scripts/fdcc/index.cfm?set=FoodSubstances>

